

Additional Results and Implementation for the paper Integrating Semantic Attributes into Deep Learning for Enhanced SAR Imagery Analysis

1 SAR Data Generation Using GAN Models

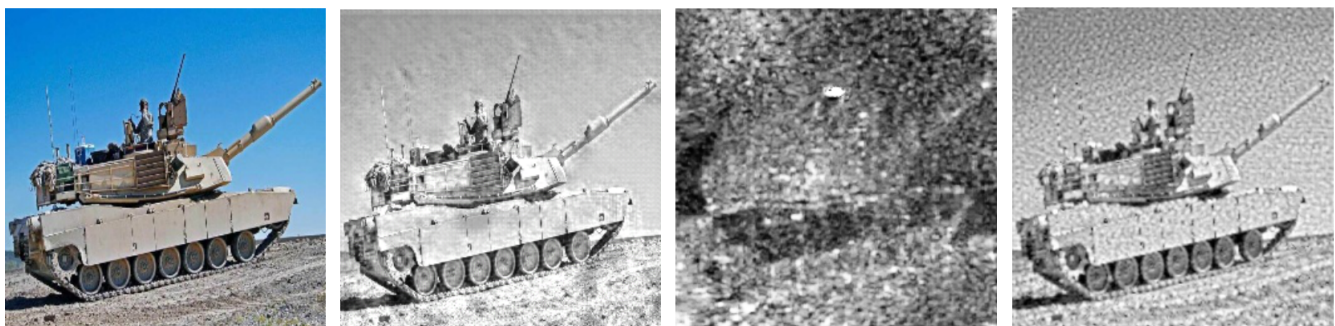
While SAR images for remote sensing applications are available, obtaining ground truth SAR data for military vehicles is challenging due to the limited availability of public data. To address this issue, we employed synthetic image generation models to generate SAR images from optical (RGB) images. We experimented with three state-of-the-art (SOTA) image-to-image translation models:

- *AttentionGAN* [7]: An advanced GAN that integrates attention modules into both the generator and the discriminator, allowing the model to focus on critical regions in the image. This allows the network to learn fine-grained mappings, thereby improving the quality, texture consistency, and level of detail of the generated images.
- *Pix2Pix* [4]: A conditional GAN framework designed for paired image-to-image translation tasks, leveraging pixel-level supervision to generate high-quality images.
- *CycleGAN* [10]: A model capable of performing unpaired image-to-image translation using cycle consistency loss, enabling the transformation of images from one domain to another without direct correspondence.

These models were trained and tested using a SAR and RGB image pair dataset from Sentinel-1 and Sentinel-2 satellites [6]. The dataset provided a diverse range of images contributing to robust model training and evaluation. Figure 1 shows an example output from each model.

To assess the performance of each model, we used the following image similarity metrics:

- *Structural Similarity Index Measure* (SSIM) [8]: SSIM measures the perceptual similarity between two images by considering changes in structural information, luminance, and contrast.
- *Peak Signal-to-Noise Ratio* (PSNR) [3]: PSNR compares the maximum possible pixel value of the image to the power of the noise (difference) between the original and generated images.
- *Learned Perceptual Image Patch Similarity* (LPIPS) [9]: LPIPS measures the perceptual similarity between two images. It is designed to better align with human perception than traditional metrics like SSIM and PSNR.



(a) RGB image from the dataset (b) SAR image generated by AttentionGAN model (c) SAR image generated by Pix2Pix model (d) SAR image generated by CycleGAN model

Figure 1: Synthetic SAR image data generation by different models. Here (a) is the input RGB image. (b), (c) and (d) are the outputs of *AttentionGAN*, *Pix2Pix*, and *CycleGAN* respectively.

- Histogram Comparison [1]: Compares pixel intensity/color histograms using methods such as correlation, Chi-square, or intersection.

Table 1: Quantitative comparison of SAR image generation models. Higher SSIM, PSNR, and histogram similarity are better, while lower LPIPS is better.

Model	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	Hist. Similarity $\uparrow$
Pix2Pix	0.060	10.81	0.387	0.934
AttentionGAN	0.046	11.01	0.390	0.961
CycleGAN	0.064	11.40	0.364	0.951

We compared Pix2Pix, AttentionGAN, and CycleGAN using SSIM, PSNR, LPIPS, and histogram similarity. Because optical-to-SAR translation involves substantial cross-modal differences, the absolute SSIM and PSNR values are low and should be interpreted only as relative comparisons among the three models. CycleGAN achieved the strongest overall quantitative performance, with the highest SSIM and PSNR and the lowest LPIPS, while AttentionGAN achieved the highest histogram similarity. Based on the combined quantitative results and qualitative inspection, CycleGAN was selected for generating the SAR-like dataset. This approach effectively bridges the gap caused by the lack of ground truth SAR data.

Shape Context Extraction

We also experimented with shape information in the form of the *Shape Context* (SC) feature descriptor, introduced by Belongie et al. [2], as additional information to fuse with the attribute and image features. The SC descriptor represents an object’s shape by placing a set of points on its boundary and computing a log-polar histogram of the relative positions of other points. Figure 2 shows the edge points of a military tank sampled by the SC descriptor.



(a) SAR image of a tank

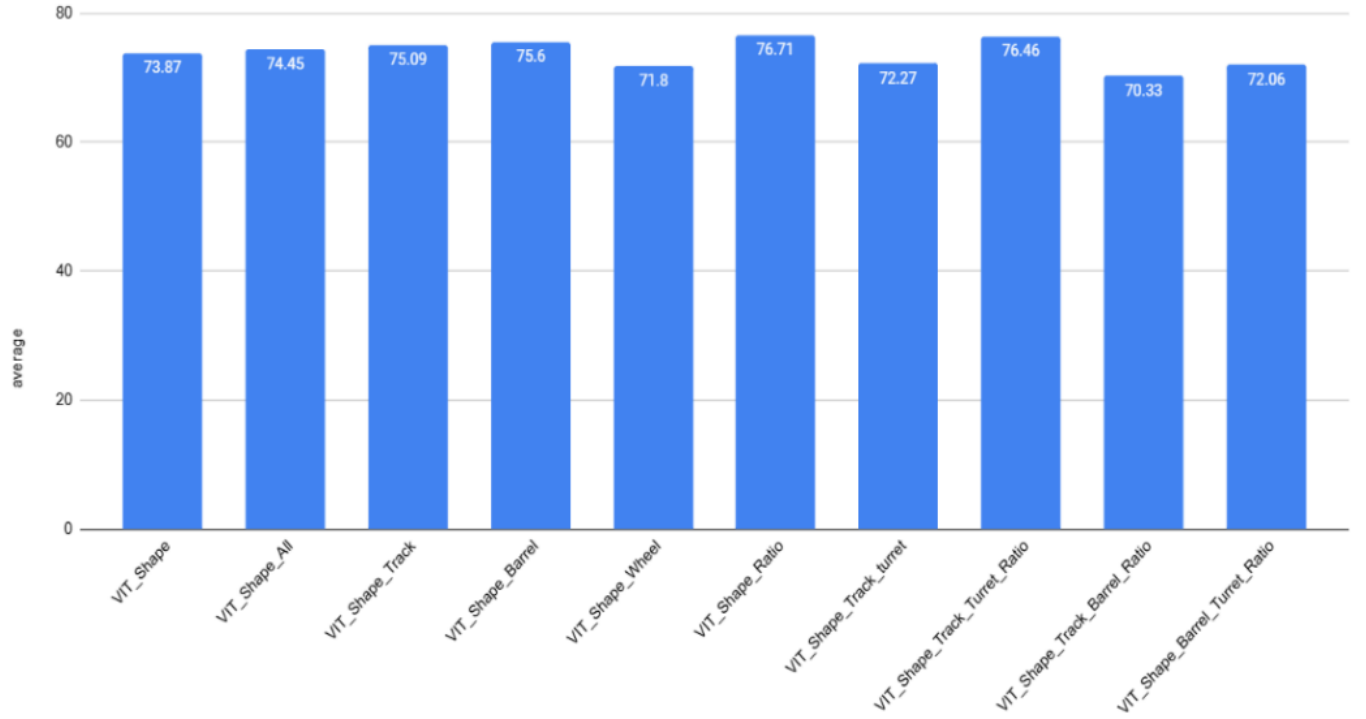


(b) Edge points placed by the SC descriptor

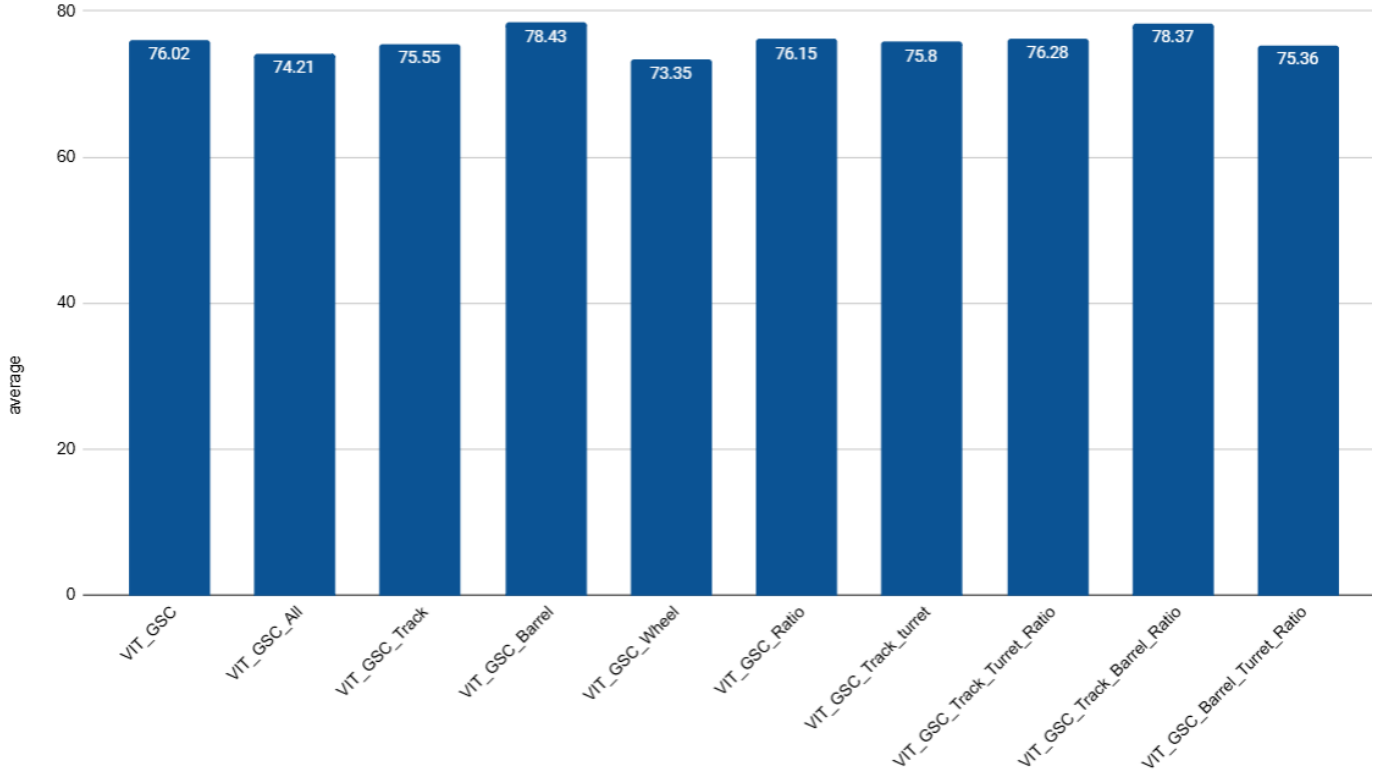
Figure 2: Getting shape information using SC feature descriptor

We store the SC information as a numerical vector that encodes the shape of an object. We also explored an improved variant of SC known as generalized shape context (GSC) [5]. GSC offers a shape descriptor that includes edge orientation details, incorporating tangent vectors alongside histogram-based information. This combination provides a more comprehensive representation of shape information, making GSC potentially more effective than the traditional SC. We performed thorough experiments with different combinations of attributes coupled with the SC and GSC. The primary objective was to determine whether adding the SC/GSC descriptor could improve baseline performance.

From Figure 3, we see that in the context of image classification using SAR data, the addition of SC/GSC information to the image features did not result in improved performance for any of the models. Although we get better results while incorporating GSC when we are not using any other attribute information, combining attributes and the GSC information results in poor performance by the model. All attributes combined (VIT_All) without any SC/GSC



(a) Comparison of VIT models' performance with different combinations of attributes and SC.



(b) Comparison of VIT models' performance with different combinations of attributes and GSC.

Figure 3: ViT performance analysis with SC and GSC

Track	Track_Barrel	Barrel_Wheel	Track_Barrel_Turret	Track_Wheel_Ratio	Track_Barrel_Turret_Wheel
Barrel	Track_Turret	Barrel_Ratio	Track_Barrel_Wheel	Barrel_Turret_Wheel	Track_Barrel_Turret_Ratio
Turret	Track_Wheel	Turret_Wheel	Track_Barrel_Ratio	Barrel_Turret_Ratio	Track_Barrel_Wheel_Ratio
Wheel	Track_Ratio	Turret_Ratio	Track_Turret_Wheel	Barrel_Wheel_Ratio	Track_Turret_Wheel_Ratio
Ratio	Barrel_Turret	Wheel_Ratio	Track_Turret_Ratio	Turret_Wheel_Ratio	Barrel_Turret_Wheel_Ratio
Pure Model	All Attributes				

Table 2: List of different combinations of attributes

information provide the most significant performance boost (79.49%), emphasizing the value of multi-attribute integration. We get better results from the model while adding a few different combinations of attributes with the SC/GSC. Still, it does not exceed the performance of the VIT_All model where we are not using any SC/GSC information.

2 Feature Fusion and Classification

In the proposed AIV-Net framework we experimented with multiple vision models, including *ConViT*, *CvT*, *ViT*, and *ConvNeXt* to extract the image features. During training, the dataset is split as 80% training and 20% validation to ensure robust evaluation. Each data sample consists of an image representing a military vehicle, different combinations of attributes, and a 6000-dimensional SC (or 12000-dimensional GSC) vector. A key detail here is we experimented with 32 different combinations of the six generated attributes so that we can compare the contribution of each of these attributes. Table 2 shows the list of these combinations of attributes. The images are then loaded and resized to 224×224 to match the input dimensions of the pretrained models.

To incorporate additional structured information, the models integrate three distinct feature representations (two when we exclude the SC/GSC representation) :

- *Image Features*: Each image is passed through the selected model (*ViT*, *CvT*, *ConViT*, or *ConvNeXt*) to extract deep feature representations.
- *Attribute Embeddings*: a linear layer that encodes the selected attributes into a 128-dimensional representation
- *SC/GSC Embeddings*: a fully connected layer encoding the SC/GSC vector into a 512-dimensional representation.

We conducted experiments with various dimensional representations and achieved the best results using the selected configurations. The mathematical representation of the feature integration process is shown in Equation (1):

$$F = [F_{\text{img}} \oplus F_{\text{attr}} \oplus F_{\text{shape}}] \quad (1)$$

F_{img} represents the image features extracted from *ViT*, *CvT*, *ConViT*, or *ConvNeXt*. $F_{\text{attr}} \in R^{128}$ is the attribute embedding encoded by a linear layer. $F_{\text{shape}} \in R^{512}$ is the SC/GSC embedding obtained via a fully connected layer. $\oplus$ denotes the concatenation of feature vectors.

$$F = [F_{\text{img}} \oplus F_{\text{attr}}] \quad (2)$$

Equation (2) shows the feature integration process when we are not integrating shape information.

$$\hat{y} = \mathcal{C}(F) \quad (3)$$

The concatenated feature vector is then passed through a fully connected classifier for final prediction as shown in Equation (3).

Model Comparison

fig. 4 illustrates the average performance of different models in the AIV-Net framework, with each bar representing a specific model configuration. The x -axis lists various models and their configurations, while the y -axis indicates the average performance (accuracy) score over all the stressors and stress levels. Models labeled as *_All* include the six attributes (*track*, *turret*, *barrel*, *wheel*, *ratio*, and *viewpoint*), while *_Pure* models do not.

For almost all the models, the *_All* versions outperform the *_Pure* versions, indicating that incorporating additional attributes improves performance. For example, *ViT*: 79.49 vs. 74.5, *CvT*: 69.8 vs. 65.88, *ConvNeXt*: 83.55 vs. 82.41.

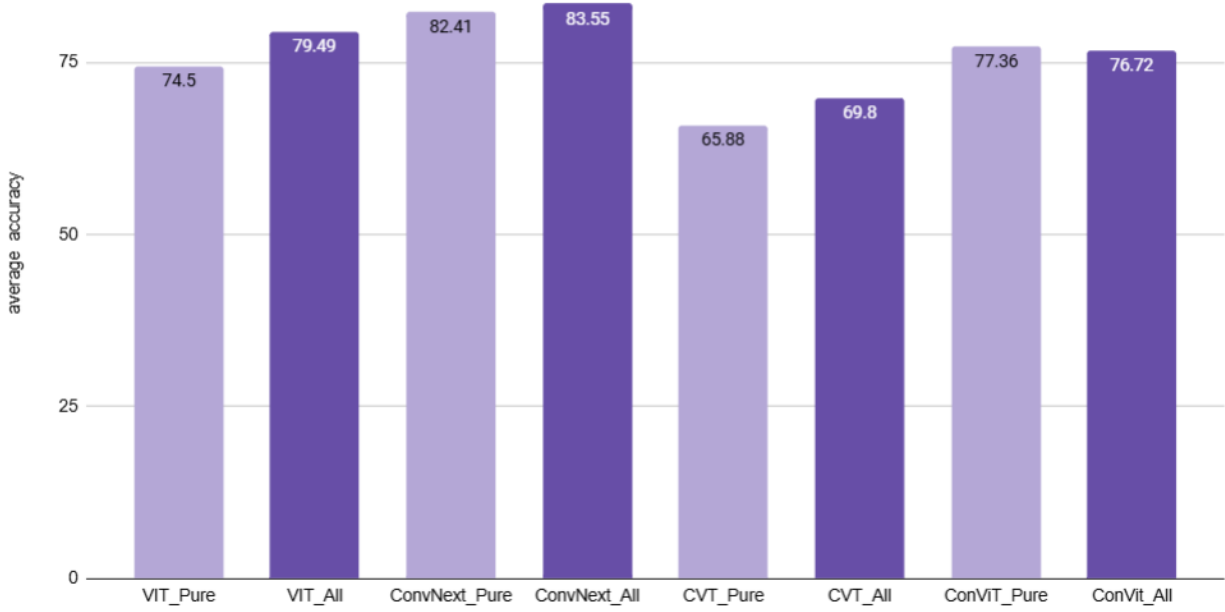


Figure 4: Comparison among different models in the AIV-Net framework on SAR data.

ConvNeXt demonstrates the highest average performance. It shows outstanding performance even without additional attributes, indicating its strong baseline capability. *CvT* has the lowest performance, with *CvT_Pure* at 65.88 and *CvT_All* at 69.8, suggesting it benefits significantly from additional attributes but still lags behind other models. *ViT* and *ConvIT* show moderate performance levels. The marginal difference between *ConViT_All* and *ConvIT_Pure* suggests that additional attributes may have a limited impact on *ConViT*.

References

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